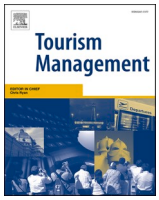
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Optimizing promotion effectiveness through timing: Leveraging promotion types aligned with temporal landmarks

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ABSTRACT

Timing is powerful in the effectiveness of promotion. This study draws on the dual processing theory to analyze how promotion types influence tourists' visit intentions to a destination from the perspective of temporal landmarks using eye tracking, a field study, and online experiments. Results suggest that matching price discount promotions with start temporal landmarks and freebie promotions with end temporal landmarks can elicit stronger visit intentions. This congruity effect is driven by cognitive and affective values. This study contributes to the research on promotion types and temporal landmarks by demonstrating their interactive effects on tourists' visit intentions. It also offers valuable insights for tourism marketers to optimize promotional effectiveness through timing.

1. Introduction

Promotions exert a significant influence on sales within the travel industry, with tourism product suppliers, such as airlines (Mason & Barker, 1996; Thiel, 1981), hotels (Lim & Ok, 2022; Ma & Li, 2023; Yang & Mattila, 2020), and travel agents (Park & Jang, 2018), frequently relying on them. This is evident from Expedia, Inc.'s 2019 report, which revealed that selling and marketing expenses accounted for over 50 % of total revenue, highlighting the importance of effective promotional strategies (myMarketing, 2024). These strategies can take various forms, including price discount promotions, bonus promotions (Byun & Jang, 2015), bundle promotions (Song et al., 2023), freebie promotions (Pan & Ha, 2022), and other patterns. Among them, two common types are price discount promotions, which offer consumers a reduction in the product price (Zoellner & Schaefers, 2015), and freebie promotions (also referred to as "free gift" or "premium" promotions), which provide consumers with a free product as an incentive for purchasing a focal product (Chandran & Morwitz, 2006; D'Astous & Jacob, 2002; Liu et al., 2022). For instance, the former is used by Contiki for deals, Trip.com for passes, and Booking.com for hotel bookings. In contrast, Tours4fun employs the latter in its travel packages, offering complimentary tickets and airport pick-up services. However, despite the widespread use of both these promotional strategies, research on their effectiveness remains scarce.

While the comparative effectiveness of price discount promotions and freebie promotions has been explored in marketing literature, their role as destination marketing strategies in the unique context of tourism marketing requires deeper examination. Unlike traditional products, tourism products are experiential (Smith, 1994). Tourists often remain uncertain about the quality of their holiday until they actually experience it (Smallman & Moore, 2010). Consequently, due to the perceived high risk and uncertainty associated with tourism consumption, tourists tend to be more sensitive to the information provided by tourism suppliers (Liu et al., 2023), making promotions particularly impactful (Wei et al., 2024). Moreover, consumers typically go through the stages of search, purchase, use, evaluate, and disposal (Moutinho, 1987). However, tourist decision-making is more complex (Smallman & Moore, 2010), as it is constrained by factors such as disposable time, budget, and family obligations (Mansfeld, 1992), and influenced by pull factors in destination choice (Kim et al., 2003). In this context, promotional strategies, such as price discounts and freebies, play a crucial role in shaping tourists' choices. Price discount promotions reduce the selling price, thereby lowering the cost, while freebie promotions enhance the product's value (Zoellner & Schaefers, 2015). A trip typically combines

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services such as transportation, accommodations, meals, and attraction fees (Wei et al., 2024). When offered as freebies, these enhance the travel experience and add value to the trip. Given the significant impact of these promotional strategies on tourists' decision-making, identifying the more effective type of promotion is a key concern for tourism marketers.

Previous literature comparing price discount promotions and freebie promotions has yielded mixed results, with some studies indicating stronger effects of price discounts and others suggesting that these discounts are less effective than freebies (e.g., Chandran & Morwitz, 2006; Darke & Chung, 2005; Foubert et al., 2018). Scholars have attempted to explain the variation in the comparative effectiveness of these promotions through product-related factors (e.g., Chandran & Morwitz, 2006), brand-related factors (e.g., Foubert et al., 2018), and individual-related factors (e.g., Crespo-Almendros et al., 2023; Pan & Ha, 2022; Weisstein et al., 2013). However, few studies have focused on contextual factors. In tourism, many choices are based on contextual "facts" (Smallman & Moore, 2010). Unlike traditional products, tourism product promotion is highly dependent on timing (Sharma et al., 2025). Tourism demand and behavior fluctuate with seasons, holidays, and weekends (Rosselló & Sansó, 2017). In addition, the decision-making process is dynamic, alternating between System 1 and System 2 processing at different phases (McCabe et al., 2016). In practice, promotional activities conducted at specific time points, such as the Mayor of London's "Let's Do Fridays" (Greater London Authority, 2024) and Trip.com's "Travel to Macau! Grab hotel deals at 10 a.m. on Monday!" with Macau hotels offering discounts starting from 45 % off (Trip.com, n.d.). These examples underscore the significance of timing in promotions, suggesting that promotional strategies may vary at different time points, which align with the concept of temporal landmarks (See Web Appendix A for more examples).

Temporal landmarks, which are employed as significant and readily accessible strategies for marketers to promote their products, denote distinctive moments or events that are prominently marked in social timeframes or personal histories (Bi et al., 2021; Dai et al., 2014; Shum, 1998). Notably, temporal landmarks such as the start (e.g., January 1) and end (e.g., December 31) of time periods are common occurrences that significantly shape people's perceptions and behaviors and are readily accessible, convenient, and effective for marketers to leverage (Bi et al., 2021; Chen & Wei, 2023). This also holds true for tourism marketers (Meng et al., 2024). Accordingly, we incorporate them into tourism promotions, aligning them with promotion types and proposing that such congruences enhance visit intentions (Su et al., 2024). Despite the practice of start and end temporal landmark strategies in tourism promotions, empirical research examining their role as effective timing cues for promotions and whether it influences their effectiveness remains scarce.

To bridge this knowledge gap, this study explores the connection between temporal landmarks and promotion types in influencing destination visit intentions. Drawing upon dual processing theory, this study hypothesizes that start temporal landmarks, when System 2 is engaged, would be more suited to a price discount promotion that requires cognitive involvement. Meanwhile, end temporal landmarks, when System 1 is engaged, would be more suited to a freebie promotion that involves affective involvement. Significantly, we discerned the underlying mechanisms of perceived value, encompassing both the cognitive and affective dimensions that influence how promotion types and temporal landmarks interact to affect tourists' intentions to visit. By examining how different types of promotions influence tourist visit intentions based on timing, this study not only advances the academic understanding of the interplay between promotion timing and type but also offers actionable insights for tourism marketers seeking to optimize their promotional strategies.

2. Literature review

2.1. Promotion: price discounts versus freebies

Sales promotion, a fundamental marketing strategy adopted by numerous companies, constitutes a substantial portion of marketing communication budgets (Hardesty & Bearden, 2003). Across industries, the effectiveness of promotions often hinges on their type, and research in the tourism context has shown that promotion type (e.g., price discount vs. free offers, price discount vs. bonus, price discount vs. gift, percentage-off discount vs. free surcharge, add-on discount vs. bundle discount, discount on the low vs. high-heterogeneity component) can shape individuals' cognitive (e.g., perceived savings, perceived quality) and affective responses, which in turn influence their behavioral intentions (Byun & Jang, 2015; Crespo-Almendros et al., 2023; Lim & Ok, 2022; Pan & Ha, 2022; Qiu et al., 2022; Song et al., 2023; Wei et al., 2024). Given their frequent use in the tourism industry, this study focuses on price discounts and freebies.

Existing research has compared price discounts and freebies, providing insights into their effectiveness from various psychological perspectives. The evidence suggests that discounts may be more effective as they are less likely to be perceived as blunt marketing gimmicks and provide greater flexibility in the allocation of savings (Foubert et al., 2018; Weisstein et al., 2013). For premium products, attribution theory and the covariation principle suggest that this promotion is more common and, therefore, less detrimental to brand perception (Zoellner & Schaefers, 2015). Conversely, other studies indicate that freebie offers may outperform discount offers. For instance, according to attribute framing, when consumers infer the quality of a target item, freebies direct their attention to the original price, thereby reinforcing their perception of the item's quality and value (Darke & Chung, 2005). Furthermore, prospect theory suggests that free promotions, processed independent of price, are more salient and focal, potentially having a stronger influence on purchase decisions (see Chandran & Morwitz, 2006). To summarize, price discounts and freebies have different characteristics, thereby affecting perceptions and behavior.

Considering the distinct advantages of price discounts and freebies, previous research has attempted to explain the variation in their comparative effectiveness through various factors, including product-related (Chandran & Morwitz, 2006); brand-related aspects (Foubert et al., 2018); and individual-related (Crespo-Almendros & Del Barrio-García, 2014; Crespo-Almendros et al., 2023; Pan & Ha, 2022; Weisstein et al., 2013). However, research on contextual factors remains insufficient. This study identifies this research gap through a comprehensive literature review (see Table 1). Contextual factors in tourism promotion are particularly important, particularly time-related factors, as evidenced by the seasonality of tourism demand (Rosselló & Sansó, 2017). Given the importance of timing in tourism, and consequently in its promotional effectiveness (Qiu et al., 2022), it is essential for tourism managers to understand how timing influences the comparative effectiveness of price discounts and freebies. To date, little attention has been devoted to this issue. The present research introduces temporal landmarks, one of the most prevalent time cues, to address this gap.

2.2. Temporal landmarks

Temporal landmarks, defined as special periods that "emerge from and stand in marked contrast to the seemingly unending stream of trivial and ordinary occurrences that happen to us everyday" (Shum, 1998), serve to provide a sense of direction, mark turning points, and establish time period boundaries (Peetz & Wilson, 2014). Researchers have classified temporal landmarks into two categories: public and personal, as well as start and end (see Yan et al., 2023). The former category is based on events; public landmarks involve significant public events, such as national elections, whereas personal landmarks are related to major life events, such as birthdays or weddings (Dai & Li, 2019). The

Table 1
Research on the effectiveness of price discounts vs. freebies.

Topic	Promotion type	Factor			Preference: discount vs. freebie	Reference
		Category	Type	Condition		
Promotional frame, quality perception, and deal value	• Discount • Every-day-low price • Free gift	–	–	–	Freebie	Darke and Chung (2005)
Promotions and negative contextual information on quality	• Monetary price promotion • Free promotion	Product	• Quality information	Negative	Freebie	Chandran and Morwitz (2006)
Price framing and online dynamic pricing	• No framing • $ off • % off • Free gift • Gift card	Individual	• Customer segments	Prospective Regular	Discount Indifferent	Weisstein et al. (2013)
Online promotion and Internet-user recall	• Monetary (price discount) • Non-monetary (free gift)	–	–	–	Product recall: discount Brand and product recall: freebie	Crespo-Almendros and Del Barrio-García (2014)
		Individual	• Attention type	Active	Overall recall: discount	
Promotions, premium product sales and brands perception	• Direct discount • Free-gift promotion • Capture/loyalty promotion • Trade-in promotion	–	–	–	Discount	Zoellner and Schaefers (2015)
The purchase effects of premiums and price cuts	• Premium • Price cut	– Brand	– • Brand characteristic	– Private label National brand	Discount Indifferent Discount	Foubert et al. (2018)
Promotion and purchasing occasions	• Price discount • Free delivery • Gift	Individual	• Attitudes toward selecting a restaurant brand	Positive	Repurchase intentions: freebie	Pan and Ha (2022)
Promotion, heritage brand equity, and power distance	• Non-monetary sales • Monetary sales	– Individual	– • Power distance	– High Low	Freebie Indifferent Freebie	Crespo-Almendros et al. (2023)
Promotion, timing, and intention to visit a destination	• Price discount promotion • Freebie promotion	Context	• Temporal landmarks	Start End	Discount Freebie	The present research

latter is based on transition points of social timetables that signal the start or end of a period (Bi et al., 2021). Given the prevalence of start and end temporal landmarks and the ease of manipulating socially constructed calendar time points in the marketplace (Bi et al., 2021; Chen & Wei, 2023), the present study aims to examine their influence.

Research has demonstrated that start and end temporal landmarks influence individuals' behavioral patterns through psychological processes (for a summary of key research, see Table 2), such as mental accounting (Shah & Li, 2024) and the spatial metaphor of time (Bi et al., 2021). Importantly, research identifies psychological resources as the mechanism through which temporal landmarks exert their influence, suggesting that these resources diminish continuously over time. When considered on a daily scale, the beginning of the day is typically associated with ample psychological resources, making this time point conducive to activities requiring cognitive engagement, as demonstrated by the *morning morality effect* (Kouchaki & Smith, 2014). In contrast, the end of the day is linked to a scarcity of psychological resources, resulting in greater susceptibility to affective product features (Bruyneel et al., 2006), such as purchasing unhealthy products and engaging with vice tweets in the evening (Yang et al., 2022; Zor et al., 2022). Building on this framework, researchers have extended the analysis to broader time scales, such as work weeks and months (Chen et al., 2023; Chen & Wei, 2023).

In the tourism domain, Meng et al. (2024) further developed this framework by examining how end temporal landmarks, as opposed to start landmarks, increased the preference for natural destinations over cultural ones among tourists, a phenomenon attributed to perceived psychological resource depletion. Additionally, Luo and Mattila (2023) explored the impact of the time of day on pleasurably painful experiences. These findings suggest that temporal landmarks shape tourism behaviors and can help guide the timing of advertising efforts to align with their psychological states. Despite this valuable insight into promotion timing, important questions remain regarding how these time-based cues can be strategically leveraged for sales promotions. Specifically, how can sales promotions align with tourists' psychological states at different temporal landmarks to maximize their effectiveness? This study seeks to fill this gap by exploring the role of temporal landmarks in sales promotion timing and their downstream effects on tourist perceptions and visit intentions.

2.3. Dual processing theory

The dual processing theory, a psychological framework, posits the existence of two different modes of processing: System 1 and System 2 (Evans, 2008; Stanovich & West, 2000). According to this theory, System 1 is characterized as being unconscious, automatic, and requires minimal effort, whereas System 2 is relatively conscious, controlled, and requires high effort (Evans, 2008). The former process is associated with affective reactions and the latter with cognition (Epstein, 1994; Shiv & Fedorikhin, 1999).

Tourists may engage both System 1 and 2 in their decision-making processes, from the initial decision to take a vacation to the selection of a destination, tour package, or mode of transport (McCabe et al., 2016). This framework has been applied to study visit intentions in

Table 2
Research on start and end temporal landmarks.

Topic	Focus	Variable	Main finding	Reference
Morality	Time of day and unethical behavior	• IV: time of day (morning vs. afternoon) • DV: unethical behavior • Med: moral awareness, self-control • Mod: moral disengagement	The reduction in moral awareness and self-control in the afternoon heightens the likelihood of individuals engaging in unethical behavior.	Kouchaki and Smith (2014)
	Time of day and (un) healthy product	• IV: time of day (morning vs. evening) • DV: (un)healthy product purchases • Med: self-control → visual attention	Consumers purchase more unhealthy products in the evening (vs. morning) due to decreased self-control.	Yang et al. (2022)
	Time of day and tweets engagement	• IV: time of day (morning vs. evening) • DV: tweet content (vice vs. virtue) • Med: self-control • Mod: temperature (high vs. low)	Twitter users' engagement shifts from virtue to vice content due to the decrease in self-control from morning to evening.	Zor et al. (2022)
Attention and visual preferences	Temporal landmarks and visual attention and judgments	• IV: temporal landmark (start vs. end) • DV: attitude • Med: attentional focus (left vs. right) → ease of processing • Mod: product location (left vs. right)	The match between a temporal landmark (start vs. end) and the location of a target object (left vs. right) enhances consumers' preference for the target object.	Bi et al. (2021)
	Temporal landmarks and food package	• IV: temporal landmark (start vs. end) • DV: food package preference (simple vs. complex) • Med: need for arousal	Priming a start (end) temporal landmark leads consumers to favor food packages with simple (complex) imagery.	Chen et al. (2023)
Lifestyle	Temporal landmarks and voluntary simplicity	• IV: temporal landmark (start vs. end) × voluntary simplicity appeal (biospheric vs. egoistic) • DV: voluntary simplicity • Med: self-transcendence, self-enhancement	Priming a temporal landmark as the start (end) of a time period increases the likelihood of consumers engaging in BVS (EVS).	Chen and Wei (2023)
Financial decision-making	End temporal landmarks and financial risk-taking	• IV: end-of-period temporal landmarks • DV: financial risk-taking • Med: optimistic belief • Mod: salience, investor experience, explicit risk reminders	End temporal landmarks enhance individuals' optimism about the potential success of their financial risks, thereby influencing their risk perceptions and propensity to take such risks.	Shah and Li (2024)
Tourism	Time of day and pleasurably painful experience	• IV: time of day (morning vs. evening) • DV: experience preference • Med: physiological arousal • Mod: ambient sensory cues, advertising sensory cues	Consumers tend to prefer less (more) pleasurably painful experiences because of the lower (higher) physiological arousal in the morning (evening).	Luo and Mattila (2023)
	Temporal landmarks and destination preference	• IV: temporal landmark (start vs. end) • DV: destination preference (natural vs. cultural) • Med: perceived psychological resource depletion • Mod: time perspective (circular vs. linear)	End (start) temporal landmarks enhance tourists' preference for natural (cultural) destinations due to perceived psychological resource depletion.	Meng et al. (2024)
	Temporal landmarks and tourism promotion effectiveness	• IV: promotion type (price discount vs. freebie) × temporal landmark (start vs. end) • DV: visit intention • Med: cognitive value, affective value	Matching a price discount with a start temporal landmark, and a freebie with an end temporal landmark, would promote tourists' visit intention.	The present research

various contexts, illustrating how both pathways simultaneously shape tourists' decision-making. For instance, in virtual tourism, mental imagery processing affects both cognition and emotion, which subsequently shape tourists' intentions to visit actual destinations (Zheng et al., 2022). Similarly, during public health crises, heuristic processing explains the direct impact of destination brand love on travel intention, while analytical processing clarifies the indirect effect mediated by resistance to negative information (Zhang & Zhu, 2024).

As previous research has shown, decision-making in tourism is not a fixed process; it is dynamic and context-dependent. At each stage, either System 1 or System 2 may be engaged, depending on various choice-related factors, including cognitive load (McCabe et al., 2016). That is, the dominant system may shift based on perceived cognitive load, which may vary across individuals. When processing resources are scarce (abundant), individuals are likely to be influenced primarily by affective reactions (cognitive reasoning) (Shiv & Fedorikhin, 1999). Research indicates that psychological resources are relatively abundant at the start of a period but diminish continuously, becoming scarce by the end (Chen et al., 2023; Chen & Wei, 2023; Meng et al., 2024). Based on this, it can be inferred that the end and start temporal landmarks may influence the activation of Systems 1 and 2 through their effects on psychological resource availability. At the start temporal landmark, individuals are more likely to engage in System 2 because of the availability of sufficient psychological resources, enabling deliberate and cognitively driven decision making. Conversely, at the end temporal landmark, resource depletion may prompt individuals to rely on System 1, which is characterized by affective and automatic responses. This study posits that temporal landmarks influence the activation of Systems 1 and 2, thereby aligning with the respective characteristics of freebies and price discounts.

2.4. Congruity effect between promotion types and temporal landmarks

Tourism marketers engage in various marketing strategies (e.g., advertising and promotion) to enhance tourists' destination visit intentions (Li & Ma, 2024; Liu et al., 2023; Su et al., 2024; Volgger et al., 2021). A key mechanism underlying the effectiveness of these strategies is congruity, which explains how tourists process promotional messages and ultimately shape their destination visit intentions. According to Osgood and Tannenbaum (1955), congruity occurs when a message connects two or more objects, influencing how they are judged. The principle of congruity in human thinking states that evaluations shift

toward alignment more closely with existing beliefs or expectations. In the context of promotional messages, the promotion type and timing are the two elements connected through the message. Based on the principle of congruity, when the promotion type aligns with the timing of the offer, consumers are more likely to perceive this alignment as consistent with their existing frame of reference, leading to a more favorable judgment. Existing literature on tourism and hospitality highlights the significance of providing the right incentive at the right time (Qiu et al., 2022), further supporting the importance of congruity between promotion type and timing in enhancing behavioral intentions.

Building on this, this study proposes a congruity effect between promotion types and their timing (i.e., temporal landmarks) in enhancing visit intentions, as explained through the lens of dual processing theory. Specifically, at the start of a period, individuals have greater psychological resources (Chen et al., 2023), which lead them to engage in System 2. In this context, attention is allocated to effortful mental activities including complex computations (Kahneman, 2011). Price discounts are typically presented as either an amount-off or a percentage-off (González et al., 2016), both of which require arithmetic calculations (Weisstein et al., 2013). Calculating the new price resulting from an amount-off promotion involves reading the regular price, noting the discount, and subtracting the discount from the regular price (DelVecchio et al., 2007). Percentage-off promotions involve an additional multiplication process (Sharifi & Aghazadeh, 2016), making such promotions relatively vague and requiring mental calculations to appreciate the total savings (Roehm & Roehm, 2011), which render them challenging to process (DelVecchio et al., 2007). Since System 2 is characterized by deliberate and analytical thinking, it is well-suited for handling the cognitive effort required for price discount promotions. We propose that at the start of a period, tourists are likely to engage in System 2, which aligns with price discount promotions, thereby leading to a stronger visit intention.

As a time period draws to a close, individuals often perceive a depletion in psychological resources (Meng et al., 2024), leading them to rely on System 1. Pan and Ha (2022) argue that free gifts are clearer and less uncertain than price discounts. This clarity makes it easier to process using System 1. Furthermore, under resource constraints, spontaneously evoked affective reactions have a greater influence on consumer behavior (Shiv & Fedorikhin, 1999), drive preferences for affective product features (Bruyneel et al., 2006). Free gifts, which are known to generate a positive affect (Suh & Yi, 2012), align with this idea. Because System 1 is characterized by automatic, affective processing, it is well suited to freebie promotions that are clear and generate a positive affect. We propose that at the end of a time period, tourists are more likely to engage in System 1, which aligns with freebie promotions, thereby fostering a stronger visit intention. Formally.

H1. A match between promotion types and temporal landmarks leads to stronger tourists' visit intentions.

H1a. At start temporal landmarks, price discount (vs. freebie) promotions lead to stronger tourists' visit intentions.

H1b. At end temporal landmarks, freebie (vs. price discount) promotions lead to stronger tourists' visit intentions.

2.5. The mediating role of cognitive and affective value

Previous studies have revealed that perceived value is correlated with visit intention and promotion. For instance, Chen and Chen (2010) demonstrated that a higher perceived value among heritage tourists correlates with more positive behavioral intentions. Kim et al. (2011) reported that food tourists' intention to revisit can be predicted using perceived value. Pandža Bajs (2015) also documented the significant influence of perceived value on tourists' future behavior. In the field of promotion, studies have identified perceived value as potential mechanism affects consumers' purchase intention, search intention and retailer reputation (Grewal et al., 1998; Sweeney et al., 1999; Yoon et al., 2014).

The existing literature categorizes perceived value into cognitive and affect-based values. Cognitive value refers to the utility obtained from the functional and economic aspects of the target, whereas affective value, arising from feelings, affective states, and the social self-concept linked to the target, denotes its emotional and social utility (Floh et al., 2014; Sweeney & Soutar, 2001). Drawing from prior research (i.e., Palazon & Delgado-Ballester, 2013; Sweeney & Soutar, 2001), this study defines cognitive value as the utility derived from a promotion because of the reduction in costs and affective value as the utility gained from the moods, feelings, and emotions evoked by a promotion.

We posit that cognitive and affective values serve as critical mechanisms linking promotion types to tourists' visit intentions across various temporal contexts. Specifically, at the beginning of the temporal period, when tourists possess more psychological resources, they engage in System 2 processing to evaluate promotions from a cognitive perspective. Tourists focus on the economic utility of promotions. Chandon et al. (2000) demonstrate that monetary promotions are perceived as offering greater savings than non-monetary promotions. Price discount promotions, which fall under the category of monetary promotions (Diamond & Johnson, 1990), involve mathematical calculations to determine the final product price (Weisstein et al., 2013) and necessitate detailed analysis and consideration. This not only engages in System 2 processing, but also enables tourists to better understand the saving benefits of the promotion. Therefore, it can be inferred that price discount promotions enhance cognitive value by highlighting the benefits of savings and ultimately fostering stronger visit intentions.

Toward the end of a temporal period, individuals tend to experience resource depletion, leading to System 1 processing, in which promotions are evaluated from an affective perspective. In this context, tourists focus on the emotional utility evoked by promotions. Freebie promotions evoke feelings of excitement and happiness (Lee & Yi, 2019). Therefore, it can be inferred that these promotions elevate perceived affective value, ultimately driving stronger visit intentions. Formally.

H2. Perceived value mediates the congruity effect between promotion types and temporal landmarks on tourists' visit intentions.

H2a. Cognitive value mediates the effect of a match between the price discount promotions and the start temporal landmarks on tourists' visit intentions.

H2b. Affective value mediates the effect of a match between the freebie promotions and the end temporal landmarks on tourists' visit intentions.

2.6. Study overview

We conducted four studies to test our hypotheses (for a summary of these studies, see Web Appendix B). Study 1A applied eye tracking techniques to objectively measure tourists' visual behavior in response to promotional advertisements. Study 1B utilized the Facebook Ads platform in a field study to investigate how temporal landmarks influenced customer responses to advertisements featuring different types of promotions. In a more controlled environment, Study 2 examined the alignment between promotion type and temporal landmarks influencing destination visit intentions. Study 3 provided evidence that tourists' cognitive and affective values drive this effect.

To demonstrate the generalizability of our findings, we used a variety of stimuli across studies. All travel packages and their prices were based on offerings from online travel agencies. The price-framing formats followed those outlined by Weisstein et al. (2013), with the discount matching the monetary value of the freebie offers. Discount rates ranged from 10 % to 40 % (Park & Jang, 2018). For freebie offers, essential components of travel packages, such as entrance fees to tourist attractions, accommodation, and transportation (Wei et al., 2024), were

selected to ensure that the total expenditure was consistent across both promotional conditions. In addition, independent pre-tests conducted for each study revealed that the participants viewed these components as essential to their trips. Even if not included in the price discount condition, it was inferred that tourists would still be willing to pay extra money. Consequently, the overall cost remained consistent, ensuring economic equivalence across promotional conditions.

3. Study 1A

This study aimed to investigate the interactive effect between promotion type and temporal landmarks, providing preliminary evidence for H1. To achieve this objective, a 2 (promotion types: price discount vs. freebie; within-subject) × 2 (temporal landmarks: start vs. end; between-subject) mixed design eye-tracking experiment was conducted. Eye-tracking technology captures viewers' eye movements and visual processing behaviors (Jia et al., 2021; Li et al., 2016), with longer viewing times indicating greater processing and interest (Zhao et al., 2022). It can be reasonably assumed that the more a participant focuses on a specific type of promotional advertisement, the more likely they are to prefer that type of promotion, thereby supporting the idea that preferences for promotions vary across different temporal landmarks.

3.1. Method

3.1.1. Stimuli materials

Dunhuang, China, was chosen as the destination, with the package price set at ¥2032 to reflect the real travel offerings on Ctrip. This price covers only the fees within Dunhuang and does not include round-trip transportation from the participants' home cities to the area. To manipulate the promotion type, entrance tickets (¥508) were selected as the freebie offer, while a 25 % discount (equivalent to ¥508) was applied for the discount promotion. Additionally, we used Photoshop to synthesize and enhance two distinct photographs of Dunhuang to avoid potential visual repetition. Consequently, four advertisements were created, each combining a different promotion type with its corresponding image.

An independent pretest (n = 60) confirmed that the perceived benefit level for the discount and freebie promotions were comparable. Both promotions were deemed realistic and credible. Furthermore, the importance of entrance tickets for this tour was validated, and complimentary tickets were perceived as worthwhile (see Web Appendix E1).

3.1.2. Sample

We recruited 90 participants (M_{age} = 29.45, 58.89 % female; details are available in Web Appendix C) from a large public university in China, with an incentive of ¥20.00. All participants had normal vision and prior travel experience, and represented various occupations, including employees, self-employed individuals, students, and government officials, thereby enhancing the generalizability of the findings (Liu et al., 2023). Data were collected in a specialized eye-tracking laboratory to ensure sufficient soundproofing and uniform illumination to mitigate external interference.

The sample size was determined through a power analysis (Faul et al., 2009) to achieve statistical power (1-β = .80) for detecting medium-sized effects (f = .25) in a 2 × 2 mixed-design ANOVA. The analysis indicated that at least 34 participants were required; the final sample exceeded this threshold, ensuring robust statistical results.

3.1.3. Equipment and procedures

Eyelink 1000 was employed as the eye-tracking equipment and configured in head-stabilized desktop mode with both a display PC and a host PC. Although viewing was binocular, only data from the right eye were captured.

This study was conducted on a Monday or a Friday, which are commonly regarded as the start and end temporal landmarks in existing studies (Chen et al., 2023; Meng et al., 2024; Shah & Li, 2024). The participants were individually admitted to the laboratory and seated in front of a monitor, with each attending only on their assigned day (N_{start} = 45, N_{end} = 45). The distance between the participants' eyes and the monitor was set at 1.75 times the width of the monitor, and the distance from the Eyelink camera to their eyes was approximately 50 cm. They were first informed that the experiment consisted of two parts: the first part involved viewing a set of advertising pictures, and the second part required them to respond to several questions. Combining eye-tracking with methods such as interviews and questionnaires allows researchers to better interpret scan patterns and reveal the underlying mechanisms of attention (see Li et al., 2020).

The data collection procedure followed these steps. First, participants were instructed to rest their chins on the designated supports mounted on the desk to stabilize their heads while their eye movements were tracked. Subsequently, the right eye of each participant was calibrated and validated. If the calibration grid did not appear symmetrical or if the validation was poor, recalibration was necessary. After validation, participants were presented with an onscreen inquiry about the day of the week and asked to indicate whether they perceived it as the start or end of the time period (Bi et al., 2021). They also reported their perceptions of the uniqueness and importance of the day (Li & Yu, 2020; Yan et al., 2023). The participants read the experimental instructions on the monitor and viewed two travel package advertisements, each featuring distinct promotion types and images. To mitigate the effects of sequence bias, a Latin square design was employed to counterbalance promotion types, advertising images, and their left-right positions. This approach generated four sets of advertisements for each temporal landmark, ensuring balance and reducing positional and visual preference biases.

Each participant viewed one set of advertisements, which were displayed for 30,000 ms. Participants were then asked to select a travel package. Participants subsequently rated the extent to which the promotion was perceived as a percentage-off discount/free gift (1 = not at all; 7 = very much). Finally, as a control for confounding variables, participants indicated their familiarity with Dunhuang and the frequency of their visits (measurement details are provided in Web Appendix D). After leaving the eye tracking device, the participants reported their demographic details.

3.2. Results

3.2.1. Manipulation checks

These findings confirmed the success of our manipulations. Participants tended to view Monday as the beginning of the workweek (M = 2.58, SD = 3.99) and Friday as the end (M = 93.33, SD = 10.10, t (88) = −56.07, $p < .001$). Both days were perceived as unique (M_{Monday} = 5.04, SD = 1.40; M_{Friday} = 5.13, SD = 1.44, t (88) = −.30, $p > .05$) and important (M_{Monday} = 5.09, SD = 1.72; M_{Friday} = 5.22, SD = 1.52, t (88) = −.39, $p > .05$). Additionally, they were highly aware of the type of promotion they received [price discount: $M = 4.98 > 4$, SD = 1.57, t (44) = 4.17, $p < .001$; freebie: $M = 5.29 > 4$, SD = 1.24, t (44) = 6.99, $p < .001$].

3.2.2. Eye tracking data

To more effectively assess the differences in visual attention between the two advertisements in each group, we created two areas of interest (AOI). Fixation duration (ms) and fixation count were used as eye-movement indicators to generate heat maps (see Web Appendix E1). Subsequently, a mixed-design ANOVA of fixation duration (F (1,88) = 17.04, $p < .001$) and fixation count (F (1,88) = 95.44, $p < .001$) revealed an interactive effect of promotion type and temporal landmarks. As shown in Fig. 1, for the fixation duration, in the start condition, participants spent more time viewing the advertisement featuring a price discount (vs. freebie) promotion message ($M_{price\ discount}$ = 12608.16, SD = 3613.74; $M_{freebie}$ = 10661.16, SD = 2594.65, F (1,88) = 9.80, p =

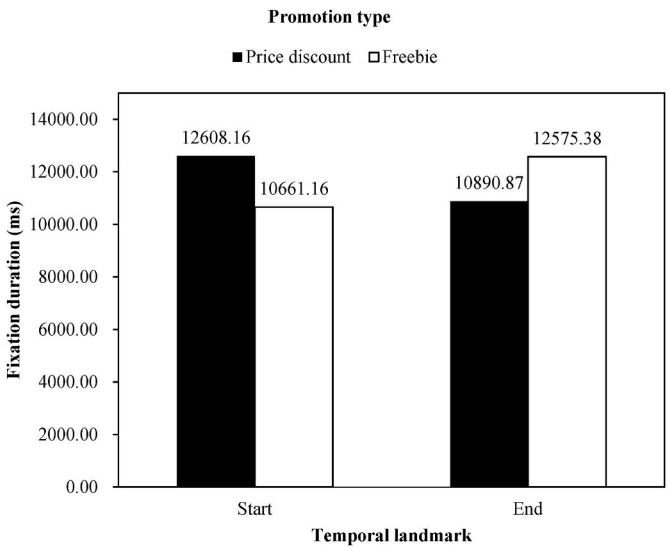


Fig. 1. Fixation duration (Study 1A).

.002). In the end condition, participants spent more time viewing advertisements featuring a freebie (vs. price discount) promotional message ($M_{\text{price discount}} = 10890.87$, SD = 2666.65; $M_{\text{freebie}} = 12575.38$, SD = 2888.54, F (1,88) = 7.33, $p = .008$). Similar results were obtained for the fixation count (start: $M_{\text{price discount}} = 65.51$, SD = 15.30, $M_{\text{freebie}} = 44.27$, SD = 12.82; F (1,88) = 56.26, $p < .001$; end: $M_{\text{price discount}} = 46.64$, SD = 9.88, $M_{\text{freebie}} = 64.53$, SD = 14.55; F (1,88) = 39.89, $p < .001$; see Fig. 2).

3.2.3. Self-report data

A chi-square test showed the main effect of temporal landmarks on promotion choice (χ^2 (1) = 7.53, $p = .006$, OR = 3.29). As expected, the majority (62.2 %) selected a travel package with a price discount on Monday, whereas 66.7 % chose a travel package with a freebie promotion on Friday. To further validate these findings, logistic regression analysis was conducted, controlling for destination familiarity and visit frequency. The results consistently support the effect ($B = -1.27$, SE = .47, Wald = 7.26, $p = .007$, OR = .28).

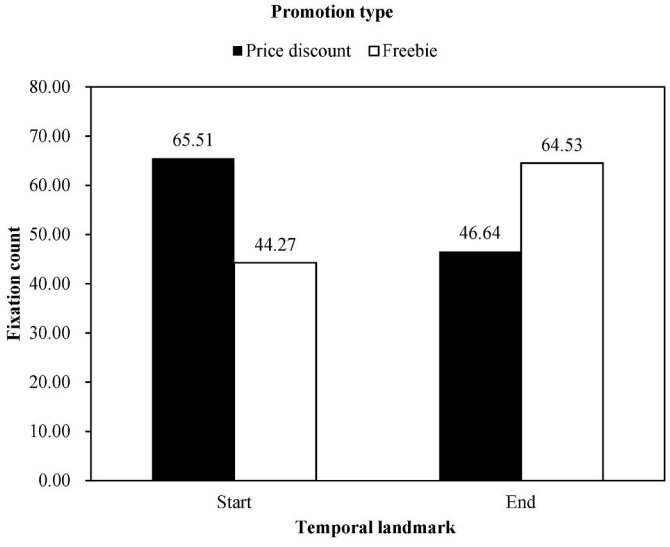


Fig. 2. Fixation count (Study 1A).

3.3. Discussion

The findings from Study 1A provided preliminary support for H1, using eye-tracking technology to examine the alignment between promotion types and temporal landmarks. Specifically, tourists paid more attention to and were more inclined to choose travel packages with price discount promotions on the first day of the workweek, while their focus shifted to freebie offers on the last day.

4. Study 1B

To enhance the external validity of the matching effect, Study 1B was conducted to examine its applicability in real-world settings. We conducted a field study on Facebook, a method recently introduced in tourism research (e.g., Lv et al., 2024; Lv et al., 2025; Meng et al., 2025), to compare advertisement versions featuring different promotional offers and determine the most effective one under start and end temporal landmark conditions. CTR (Link click-through rates) was used as a proxy for customer engagement and motivation. In this study, it represents the percentage of impressions (i.e., the number of times that your adverts were on-screen; Meta, n.d.) that received a link click to learn more about the destination out of the total number of impressions (Meng et al., 2025).

4.1. Method

4.1.1. Stimuli materials

Grand Canyon National Park, USA, was selected as the destination for a camp tour with the package price set at $400 (excluding airfares or related transportation between the home and departure city) to reflect real travel offerings on Tours4fun. To manipulate promotion types, sleeping bag rentals ($40) were selected as the freebie offer, while a 10 % discount (equivalent to $40) was applied for a discount promotion. To manipulate temporal landmarks, Monday and Friday were used as start and end temporal landmarks, respectively.

4.1.2. Pretest

To ensure the appropriateness of the manipulation of temporal landmarks and promotion types, two pre-tests were conducted using Amazon Mechanical Turk (MTurk) (see Web Appendix E2). In the first pretest, 60 participants ($M_{\text{age}} = 31.58$, 65.00 % female) were randomly assigned to either Monday or Friday and asked to identify whether they considered the day to be the beginning or conclusion of the workweek. Participants also rated the uniqueness and importance of that day. The results confirmed that Monday was perceived as the start ($M = 2.10$, SD = 2.09) and Friday as the end ($M = 92.87$, SD = 7.84, t (58) = −61.29, $p < .001$), with no significant differences in perceived uniqueness or importance.

The second pretest involved 60 participants ($M_{\text{age}} = 32.57$, 63.33 % female) assigned to either the discount or freebie promotion. Participants rated the extent to which they perceived the promotion as a percentage-off discount/free gift. Both promotion types were clearly recognized (price discount: $M = 5.30 > 4$, SD = 1.37, t (29) = 5.20, $p < .001$; freebie: $M = 5.17 > 4$, SD = 1.46, t (29) = 4.36, $p < .001$). Both types were rated equally in terms of benefits and plausibility. The importance of sleeping bags showed no significant difference between the groups, and the freebie promotion, which included sleeping bag rental, was rated as worthwhile.

4.1.3. Procedure

This study employed a 2 (promotion type: price discount vs. freebie) × 2 (temporal landmarks: start vs. end) between-subjects factorial design conducted on Facebook. Two versions of advertisements were created, with participants showing either the price discount or freebie ad on Monday (November 25, 2024, 12:00 a.m. CST – November 26, 2024, 12:00 a.m. CST) or Friday (November 29, 2024, 12:00 a.m. CST –

November 30, 2024, 12:00 a.m. CST). All other ad content remained consistent across the groups. Participants were individuals aged 18 and older from the Central United States, ensuring alignment with the CST time zone.

4.2. Results

The campaign generated 31,047 impressions and 290 link clicks. At the start temporal landmark, the price discount ads received 8269 impressions and 91 link clicks, whereas freebie ads garnered 7249 impressions and 51 link clicks. At the end temporal landmarks, the price discount ad received 8071 impressions and 60 link clicks, whereas the freebie ads received 7458 impressions and 88 link clicks. A chi-square analysis revealed that at the start temporal landmarks, participants exhibited a higher likelihood of clicking on discount ads (1.10 %) than on freebie ads (.70 %, χ^2 (1) = 6.71, p = .010, OR = 1.57). Conversely, at the end temporal landmarks, participants showed a higher tendency to click on freebie ads (1.18 %) than on discount ads (.74 %, χ^2 (1) = 7.83, p = .005, OR = .63).

4.3. Discussion

The results of Study 1B showed that at start (end) temporal landmarks, the CTR was higher for the price discount (freebie) ads, providing further evidence for the matching effect between promotion types and temporal landmarks.

In conclusion, Studies 1A and 1B, using visual attention, preference, and CTR as dependent variables, supported our main proposition. Furthermore, Study 1B enhanced external validity. Nevertheless, considering the potential confounding factors in the field experiment (Chen et al., 2024), and the lack of exploration of downstream outcomes, we subsequently carried out a series of scenario-based experiments to strengthen internal validity and investigate their joint effects on visit intentions.

5. Study 2

Study 2 aimed to overcome the limitations of Study 1, which focused on matching effects but did not capture visit intentions. By adopting subjective measures, this study directly examined participants' behavioral intentions, offering supplementary support for the proposed interactive effect and enabling a comprehensive test of H1. Additionally, a control condition without promotional claims was included to confirm that the observed effects were driven by the alignment of promotion type and temporal landmarks.

5.1. Method

5.1.1. Stimuli materials

A tour from Miami to Orlando, USA was selected as the destination, with the package price set to reflect the actual travel offerings on Tours4fun. The price covers only the fees associated with travel from Miami to Orlando, and does not include transportation from the participants' home cities to Miami or the return trip from Orlando to their home cities. Two promotional strategies were tested: a freebie promotion ($1354.55, plus a complimentary $149 ticket for a seat on the Miami to Orlando "Brightline" high-speed train) and a discount promotion (was $1354.55, now 11 % off). Additionally, according to Weisstein et al. (2013), the no-framing condition served as the baseline ($1205.55).

An independent pre-test (n = 90) revealed that the perceived benefit levels for the discount and freebie groups were similar. Both promotions were considered realistic and credible. The importance of the train ticket for this tour was confirmed, and the complimentary ticket was perceived as worthwhile (see Web Appendix F).

5.1.2. Sample and procedure

We recruited 420 participants from the US via MTurk and offered them an incentive of $.27. Participants were randomly assigned to one of the six conditions in a 3 (promotion type: price discount vs. freebie vs. control) × 2 (temporal landmark: start vs. end) between-subjects design. After excluding 16 invalid responses, 404 (M_{age} = 31.13, 59.16 % female; details are available in Web Appendix C) valid responses were retained. An a priori power analysis determined that at least 211 participants were necessary to achieve, based on an ANOVA with an effect size of .25 and power = .80.

The experiment was conducted on November 30, 2024 and December 1, 2024, to manipulate the temporal landmarks. At the outset of the questionnaire, participants were first asked to record the date and indicate whether they regarded the day as the start or end of the temporal period, and to assess how unique and important they perceived the day to be. The participants were then instructed to imagine browsing the Internet when a travel package ad for the Miami to Orlando tour attracted their attention.

Following the ad exposure, participants rated their visit intentions on a three-item scale (Cronbach's α = .916) adapted from Li and Ma (2024). A manipulation check item regarding the promotion type was included in the price discount and freebie promotion groups. To control for potential confounding variables, participants reported their familiarity with the destination, frequency of visits, as well as their travel preferences, and purchase history (measurement details are provided in Web Appendix D). Finally, the participants reported their demographic information.

5.2. Results

5.2.1. Manipulation checks

Participants tended to perceive December 1 as the start of the month (M = 2.25, SD = 3.03) and November 30 as the end of the month (M = 93.43, SD = 15.35, t (402) = −82.43, p < .001). Both days were perceived as unique ($M_{\text{December 1}}$ = 5.22, SD = 1.28; $M_{\text{November 30}}$ = 5.20, SD = 1.41, t (402) = −.10, p > .05) and important ($M_{\text{December 1}}$ = 5.01, SD = 1.38; $M_{\text{November 30}}$ = 5.08, SD = 1.49, t (402) = −.51, p > .05). Additionally, participants in the promotion group recognized the type of promotion they received [price discount: M = 5.33 > 4, SD = 1.37, t (133) = 11.23, p < .001; freebie: M = 5.19 > 4, SD = 1.27, t (134) = 10.89, p < .001], indicating the success of our manipulation.

5.2.2. Visit intention

A two-way ANOVA on visit intentions indicated no significant main effect of temporal landmark (F < 1, p > .05), but a significant main effect of promotion type (F (2,398) = 27.80, p < .001) and a significant interaction effect (F (2,398) = 17.60, p < .001). As illustrated in Fig. 3, participants in the start condition demonstrated a higher intention to visit the destination in the price discount promotion group ($M_{\text{price discount}}$ = 5.56, SD = 1.23) than in the freebie (M_{freebie} = 4.62, SD = 1.57, p < .001) and control groups (M_{control} = 4.05, SD = 1.35, p < .001). Additionally, participants in the freebie group exhibited higher visit intention than those in the control group (p = .039).

Conversely, in the end condition, participants' visit intention was significantly higher for the freebie promotion group (M_{freebie} = 5.64, SD = .92), compared to those in the price discount ($M_{\text{price discount}}$ = 4.66, SD = 1.33, p < .001) and the control group (M_{control} = 4.10, SD = 1.46, p < .001). Participants in the price discount group also exhibited higher visit intention than those in the control group (p = .043). Moreover, even after controlling for destination familiarity, visit frequency, travel preference, and purchase history, the interactive effect between promotion type and temporal landmarks remained statistically significant (F (2,391) = 19.11, p < .001), thus supporting H1.

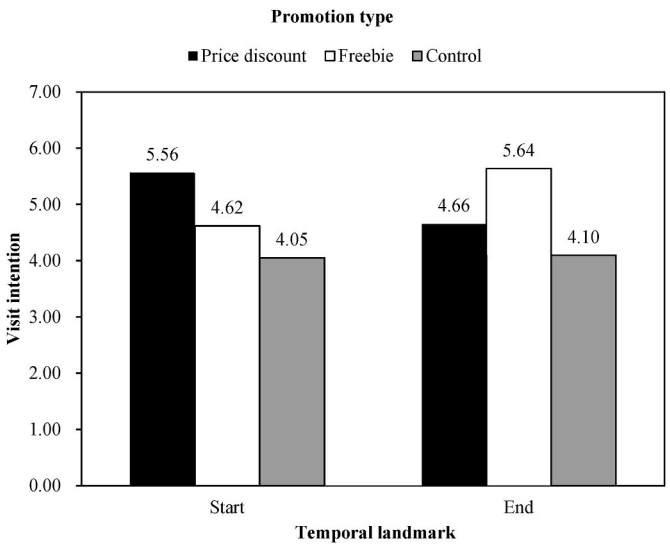


Fig. 3. Visit intention (Study 2).

5.3. Discussion

The findings from Study 2 further substantiate H1. Specifically, it was found that pairing start temporal landmarks with price discount promotions enhanced visit intention. In contrast, aligning end-temporal landmarks with freebie promotions increased visit intentions. Furthermore, the "no-framing" (control) advertisement resulted in lower visit intention compared to the promotional conditions. This effect may stem from the potential of promotions to enhance consumers' perceived value (Hardesty & Bearden, 2003). In the following study, we explored perceived value as a potential underlying mechanism.

6. Study 3

Study 3 aimed to further investigate the underlying psychological mechanisms influencing the interaction effect of promotion type and temporal landmarks on travel intention, thereby testing H2. To achieve this, subjective measures were utilized because of their strong correlation with the target variables and their ability to capture unobservable constructs (Jahedi & Méndez, 2014) such as psychological perceptions and processes. Additionally, this study addressed the potential confounding variables identified in Studies 1 and 2, in which the use of real destinations introduced uncontrolled variations.

6.1. Method

6.1.1. Stimuli materials

In this study, a fictitious destination, Tsing Lake, was created at a package price of ¥1,120, reflecting typical lake tours on Ctrip. This price covers only the fees within Tsing Lake and does not include round-trip transportation from the participants' home cities to the area. To manipulate the promotion type, a cruise ticket (¥168) served as the freebie offer, while a 15 % discount (equivalent to ¥168) was applied for a discount promotion.

An independent pre-test (n = 60) revealed no significant differences in perceived benefit levels between the discount and freebie groups. Both promotions are deemed realistic and credible. Furthermore, the importance of cruise tickets for this tour was confirmed with complimentary tickets perceived as worthwhile (see Web Appendix G).

6.1.2. Sample and procedure

A total of 500 Chinese participants were recruited through Credamo and received an incentive of ¥2.00. This study adopted a 2 (promotion types: price discount vs. freebie) × 2 (temporal landmarks: start vs. end) between-subjects factorial design. After excluding 17 invalid responses, 483 (M_{age} = 31.95, 58.59 % female; details are available in Web Appendix C) valid questionnaires were collected. An a priori power analysis indicated that a minimum of 179 participants was required to achieve sufficient power based on an ANOVA with an effect size of .25 and power of .80.

To manipulate the temporal landmark, participants were initially instructed to imagine themselves on either the first or the last day of their annual leave (Meng et al., 2024), and to indicate whether they considered the day as the beginning or the end of the temporal period. They also rated the perceived uniqueness and importance of the day. Subsequently, participants were instructed to imagine browsing the Internet when they encountered a destination advertisement for Tsing Lake.

After viewing the advertisement, participants indicated their visit intention (Cronbach's α = .863). To measure the mediator, participants completed a six-item cognitive value scale (Cronbach's α = .940) and a seven-item affective value scale (Cronbach's α = .943), adapted from Palazon and Delgado-Ballester (2013). A manipulation check item regarding promotion type was incorporated for both the price discount and freebie promotion groups. To control for confounding variables, participants reported their destination familiarity, travel preferences, and purchase history (measurement details are provided in Web Appendix D). They finally submitted their demographic information.

6.2. Results

6.2.1. Manipulation checks

Our manipulation was successful. Participants perceived the first day of annual leave as the start (M = 2.59, SD = 4.48) and the last day as the end (M = 92.40, SD = 16.71, t (481) = −80.74, p < .001). Both days were perceived as unique ($M_{\text{first day}}$ = 5.16, SD = 1.21; $M_{\text{last day}}$ = 5.20, SD = 1.09, t (481) = −.36, p > .05) and important ($M_{\text{first day}}$ = 5.12, SD = 1.31; $M_{\text{last day}}$ = 5.10, SD = 1.25, t (481) = .17, p > .05). Additionally, participants were well aware of the presented promotion [price discount: M = 5.23 > 4, SD = 1.31, t (238) = 14.52, p < .001; freebie: M = 5.22 > 4, SD = 1.41, t (243) = 13.49, p < .001].

6.2.2. Visit intention

The results of the 2 (promotion types: price discount vs. freebie) × 2 (temporal landmarks: start vs. end) ANOVA yielded no significant main effect of promotion type (F < 1, p > .05) or temporal landmark (F = 2.47,

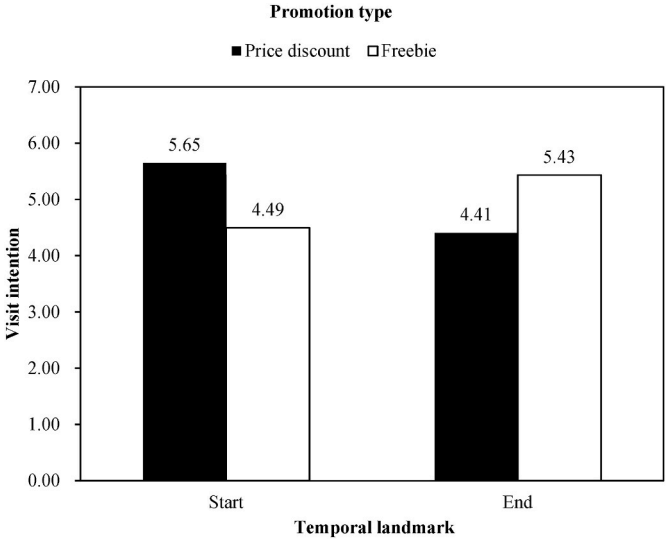


Fig. 4. Visit intention (Study 3).

$p > .05$), but revealed a significant interaction effect (F (1,479) = 129.01, $p < .001$). Fig. 4 shows that in the start condition, price discount (vs. freebie) promotion led to greater visit intentions ($M_{\text{price discount}} = 5.65$, SD = .92; $M_{\text{freebie}} = 4.49$, SD = 1.07, F (1,479) = 72.60, $p < .001$). In the end condition, freebie (versus price discount) promotion led to greater visit intention ($M_{\text{price discount}} = 4.41$, SD = 1.20; $M_{\text{freebie}} = 5.43$, SD = 1.00, F (1,479) = 56.91, $p < .001$), supporting H1. Furthermore. The interactive effect remained significant even after controlling for destination familiarity, travel preferences, and purchase history (F (1,473) = 125.17, $p < .001$).

6.2.3. Mediation analysis

Bootstrapping analyses (Model 8, 5000 samples; Hayes, 2017) were conducted to test H2. Cognitive and affective values were included in the model. In terms of cognitive value, the results indicated that in the start condition (indirect effect = −.2253, SE = .0685), the 95 % confidence interval (CI) ranged from −.3776 to −.1126, whereas in the end condition, the 95 % CI ranged from −.1844 to .0129, providing evidence that cognitive value mediated the interaction effect on tourists' visit intention only in the start temporal landmark condition. The complete path coefficients are depicted in Fig. 5.

In terms of affective value, the results indicated that in the end condition (indirect effect = .3275, SE = .0757), the 95 % CI ranged from .1851 to .4832, whereas in the start condition, the 95 % CI ranged from −.0138 to .2373, providing evidence that affective value mediated the interaction effect on tourists' visit intention only in the end temporal landmark condition. The complete path coefficients are depicted in Fig. 6.

6.3. Discussion

Study 3 further supports this interaction effect and reveals the mechanisms of cognitive and affective values. Specifically, when a temporal landmark is framed as the beginning of a temporal period, tourists perceive the price discount promotion as having higher cognitive value and exhibiting greater visit intentions; conversely, when a temporal landmark is framed as the conclusion of a temporal period, tourists perceive the freebie promotion as having higher affective value and exhibiting greater visit intentions.

7. General discussion

Previous research demonstrated the significant role of timing in the effectiveness of promotions. This study examined the interactive effect of promotion type and temporal landmarks on tourists' intentions to visit a destination. Across four studies, we demonstrated that priming start temporal landmarks resulted in greater visit intentions when destinations were promoted with price discount (vs. freebie) promotions, whereas priming end temporal landmarks led to higher visit intentions when destinations were promoted with freebie (vs. price discount) promotions (Studies 1 and 2). Our findings indicated that cognitive and affective values mediate this effect (Study 3). The robustness of these findings is underscored by the convergence of evidence derived from eye-tracking, a field study, and online experiments. These studies employ various operationalizations of discount rates, freebie offers, temporal landmarks, and samples.

7.1. Theoretical implications

First, this study contributes to a better understanding of the effectiveness of promotions. Existing studies comparing price discounts and freebies have primarily focused on internal (e.g., Crespo-Almendros et al., 2023; Pan & Ha, 2022; Weisstein et al., 2013) and product-specific factors (e.g., Chandran & Morwitz, 2006; Foubert et al., 2018). However, the roles of external factors, particularly timing, remain underexplored. In tourism, where demand fluctuates with yearly, monthly, and weekly seasonality (Rosselló & Sansó, 2017), the timing of promotions is especially crucial. This study investigated temporal landmarks as subtle yet pervasive contextual cues and examined the comparative effectiveness of price discount and freebie promotions within this framework. By advancing our knowledge of temporal cues as external stimuli, our findings address a gap in the promotion research.

Second, this study adds value to the current scholarly works on temporal landmarks by extending their application to tourism research. Although recent studies have investigated the influence of start and end temporal landmarks on consumer behavior (e.g., Bi et al., 2021; Chen et al., 2023; Chen & Wei, 2023) and financial behavior (e.g., Li & Shah, 2020; Shah & Li, 2024), the effect of these temporal landmarks on tourist responses remains underexplored. Meng et al. (2024) suggest that start and end temporal landmarks influence tourists' preferences for different types of destinations, highlighting their significance in promotion (i.e., advertising) timing. This study further advances theoretical understanding by examining how temporal landmarks can be strategically leveraged in sales promotion timing, enhancing our insight into their impact on tourist decision-making.

Third, this study advances the dual processing theory by integrating it into the context of tourism promotion, providing a novel perspective on how temporal landmarks influence the activation of Systems 1 and 2. The tourism literature has identified several factors that activate these systems simultaneously. For instance, during the formation of travel intention, destination brand love influences travel intentions through both heuristic and analytic processing (Zhang & Zhu, 2024). as well as the elaboration and quality of mental imagery processing in virtual tourism, which affects visit intentions (Zheng et al., 2022). During travel planning, systematic and heuristic cues from AI-based recommendation systems influence adoption intentions (Shi et al., 2021). This study extends these frameworks by incorporating temporal landmarks as key factors that influence the dominance of either System 1 or System 2. At the beginning of the period when psychological resources are abundant, System 2 is more likely to be engaged, fostering greater cognitive processing and preferences for effortful promotions (e.g., price discounts). In contrast, at the end of the period, as resources are depleted, System 1 is activated, increasing the influence of affective promotions (e.g., freebies). By examining the role of temporal landmarks in promoting cognitive and affective responses, this study contributes to a deeper understanding of the dynamic nature of tourism decision-making. Moreover, it offers a valuable extension of the dual processing theory.

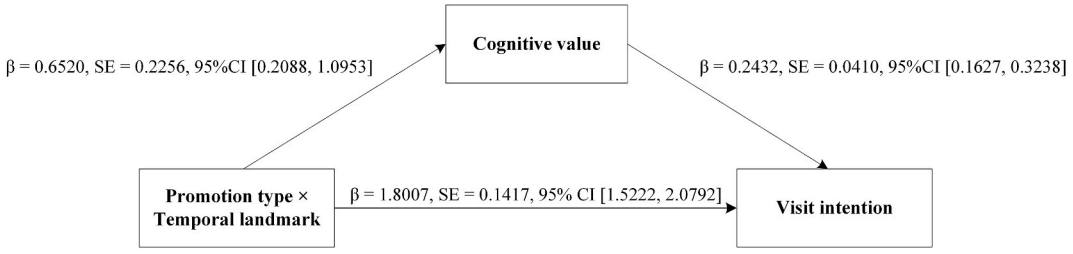


Fig. 5. The mediating role of cognitive value.

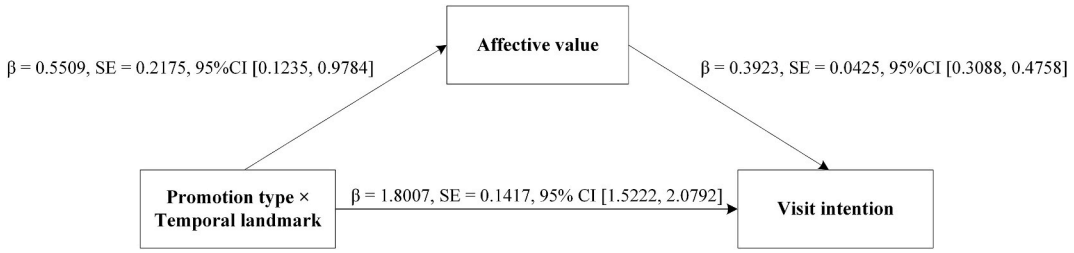


Fig. 6. The mediating role of affective value.

7.2. Managerial implications

First, the management should strategically leverage temporal landmarks to craft effective promotional advertisements. This study highlights the fact that the effectiveness of different promotions in stimulating visit intentions varies depending on the activation of specific temporal landmarks. We observed that price discount promotions elicited higher visit intentions among tourists than freebie promotions at the beginning of a time period, while the opposite effect occurred at the end. These findings hold significant practical implications for framing prices and presenting price offers in alignment with advertisement timing. For instance, in a promotional campaign for a three-day tour from Los Angeles to Las Vegas, advertisements could feature a 15 % discount at the start of the time period while emphasizing complimentary offerings—such as the Las Vegas nights tour, which includes the Downtown Fremont Street Experience, the Fountains of Bellagio, and an outside view of the Sphere—toward the end.

Second, tourism marketers can strategically leverage temporal landmarks based on the nature of their promotional offers. Price discounts may be introduced at the beginning of a temporal period, whereas freebie promotions can be positioned toward the end. Even on the same calendar day, marketers can effectively frame temporal landmarks to align them with promotional strategies. For example, a marketer promoting a Bali travel package with a 12 % discount targeting college students could frame late August as the beginning of the college semester and encourage them to plan a pre-semester getaway. Similarly, if the travel package features the complimentary Catamaran Cruise and BBQ lunch, the marketer could frame late August as the culmination of the August work period, appealing to office workers seeking to reward themselves after a month of hard work.

Third, this study demonstrated the mediating role of cognitive and affective values in the interaction effect on visit intention. At the start temporal landmark, tourists perceive price discount promotions as possessing higher cognitive value. Conversely, at the end temporal landmark, freebie promotions were perceived as having higher affective value. Keeping this in view, it is suggested that marketing managers reinforce their communication of the cognitive (e.g., emphasizing "Think WISELY about the cost and benefits,'' "PLAN your money well,'' "LOCK IN your savings") or affective value (e.g., "ENJOY free entry tickets," "Unlock the EXCITEMENT of complimentary hotel upgrades") through advertising in the campaign content, thereby stimulating tourists' intentions to visit the destination.

7.3. Limitations and future research

This study had certain limitations that offer insights for future research. First, visit intention was assessed using visual attention, CTR, and self-reported measures. Although these methods capture attention, interest, and behavioral intentions, they do not fully reflect actual behavior. Future research could incorporate objective behavioral data, such as bookings or purchases, to directly measure behavior and strengthen external validity. Furthermore, future studies could enhance the mediation effect' s external validity by incorporating physiological methods, such as electroencephalography (EEG).

Second, we only considered the effect of a match between the promotion types and temporal landmarks. Studies show that the size of an offer may influence the effectiveness of a promotion (Grewal et al., 1996; Hardesty & Bearden, 2003). Subsequent investigations should examine the role of promotional offer level. In addition, future research should assess whether the interactive effects of the promotion type and temporal landmarks are moderated by specific boundary conditions. For example, the proposed effect should be explored among individuals who are either highly or less price-conscious.

Third, this study delineates the beginning and end of the calendar periods. In business practice, midmonth and midyear promotions imply that the midpoint of the time period might hold significance. Future research should explore whether a mid-temporal landmark effect exists.

Finally, this study affirms that the interaction between promotion type and temporal landmarks in shaping visit intentions can be attributed to the cognitive and affective values perceived by tourists. Further investigations are necessary to explore other possible explanations for this observed effect.

CRediT authorship contribution statement

Chenya Ma: Writing – review & editing, Writing – original draft, Resources, Methodology, Funding acquisition, Data curation. **Xin Liu:** Writing – review & editing, Writing – original draft, Methodology. **Lu (Monroe) Meng:** Writing – review & editing, Writing – original draft, Supervision, Resources. **Yushi Jiang:** Writing – review & editing, Resources, Project administration.

Impact statement

This research delves into the underexplored intersection of temporal landmarks and promotional strategies in the travel industry, offering novel insights into the timing of promotional campaigns. By examining the differential impacts of price discount and freebie promotions in relation to start and end temporal landmarks, our study provides actionable guidance for tourism marketers. We reveal that the effectiveness of promotional types is contingent upon the temporal context, with price discount promotions being more effective at the start of a period, and freebie promotions at the end. This discovery underscores the importance of aligning promotional timing with temporal landmarks to maximize sales and visitation intentions, thus contributing a new dimension to marketing strategies in the travel sector.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tourman.2025.105198.

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